

# Yash Sukhdeve

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## Summary

Ph.D. student in Electrical and Computer Engineering with applied ML and data science experience across biomedical image classification research, computer vision pipelines, scientific Python/R workflows, and reproducible data-processing systems. Interested in applying machine learning and image-analysis workflows to multiplex immunofluorescence, spatial biomarker discovery, and translational oncology research.

## Technical Skills

**Scientific Programming:** Python, R, SQL, Pandas, NumPy, scikit-learn, PyTorch project experience, Jupyter, Git/GitHub

**ML and Image Analysis:** computer vision, object detection, transfer learning, federated learning, biomedical image classification, feature extraction, model benchmarking, error analysis, data preprocessing

**Data/Systems:** Flask, Docker, SQLite, MySQL, AWS EC2, Linux, data validation workflows, ETL-style pipelines, technical reports, Tableau

## Relevant Experience

**Research Assistant**, AVHBAC Lab, CITeR, Clarkson University Jan 2025 – Present  
*Potsdam, NY*

- Build reproducible ML research workflows for biometric AI, speaker recognition, diarization, and synthetic-media analysis, emphasizing data preprocessing, benchmarking, and careful experimental reporting.
- Analyze model behavior across datasets and document findings in technical reports; experience translates directly to validating algorithms for image-derived biomarker and spatial-analysis pipelines.

**Data Science Intern**, Clarkson University Feb 2024 – Jan 2025  
*Potsdam, NY*

- Developed and deployed a Flask-based ERP data validation platform using Python, Docker, SQLite, and MySQL to automate data quality checks and improve reliability of institutional data workflows.
- Implemented reusable validation logic and database-backed processing routines, strengthening practical skills in workflow design, debugging, and maintainable scientific/data software.

**Research Intern**, Indian Institute of Information Technology Nagpur Jan 2022 – Jun 2022  
*Nagpur, India*

- Conducted deep learning research on skin cancer classification using federated learning, with emphasis on biomedical image classification, distributed model training concepts, and privacy-aware learning.
- Reviewed 50+ papers on skin cancer classification and federated learning; synthesized model architectures, datasets, and methodological gaps into presentations for research meetings.

**Graduate Teaching Assistant**, Clarkson University Graduate School Oct 2022 – Dec 2023  
*Potsdam, NY*

- Mentored students in R, SQL, Tableau, Python, Pandas, and NumPy for information systems and big data processing coursework; explained analytical concepts and supported hands-on lab troubleshooting.

## Selected Projects

**Wild Animal Detection with MegaDetector** 2024

- Built a computer-vision workflow for image/video object detection and classification using MegaDetector, transfer learning concepts, AWS EC2, Deep Learning AML, and data-processing scripts for ingesting, processing, and storing model outputs.

**Healthcare Translation Application** 2025

- Built a Flask web application for medical conversation transcription and multilingual translation using OpenAI Whisper, GPT-3.5-turbo, JavaScript, Bootstrap, secure environment variables, and Vercel deployment.

**ERP Data Validation and Containerized Workflow Platform** 2024

- Designed a Dockerized Python/Flask/MySQL workflow for automated validation of structured records, including database checks, user-facing interfaces, local testing, and deployment-oriented configuration.

## Education

**Clarkson University** Potsdam, NY  
Ph.D., Electrical and Computer Engineering Dec 2025 – Present

M.S. pathway, Electrical and Computer Engineering Jun 2025 – Dec 2025

**Clarkson University Graduate School** Potsdam, NY  
M.S., Data Science Aug 2022 – Dec 2023

**Indian Institute of Information Technology Nagpur** Nagpur, India  
B.Tech., Computer Science 2018 – 2022

## Certifications and Publications

**Certifications:** Python for Everybody Specialization; Database Design and Basic SQL in PostgreSQL; NVIDIA DLI - Building Transformer-Based NLP Applications

**Publications:** *Descriptor: Extended-Length Audio Dataset for Synthetic Voice Detection and Speaker Recognition (ELAD-SVDSR); Crime Prediction Using K-Nearest Neighboring Algorithm*